Report 2: Attention Task Validation

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1 Introduction

1.1 Background and Motivation

Selective attention is a core cognitive function [1]. Laboratory visual-search paradigms are the gold standard for measuring it, but gamified mobile assessments offer a scalable alternative [2] that requires validation.

This report documents the statistical inference for our 2×2 mixed factorial study validating a mobile game-based modality against a laboratory modality under different target loads (single vs. multiple). We applied mixed ANOVA, non-parametric methods, and post-hoc tests to evaluate our six hypotheses.

1.2 Research Hypotheses

H1 — Concurrent Validity

H_0: There is no correlation between performance on the Game and Lab task ($r = 0$).
H_1: Performance on the Game task is positively correlated with performance on the Lab task ($r > 0$, one-tailed).

H2 — Load Effect

H_0: RT and accuracy do not differ between the Single Target and Multiple Target conditions ($\mu_{\text{single}} = \mu_{\text{multiple}}$).
H_1: The Multiple Target condition results in slower RTs and lower accuracies compared to the Single Target condition, reflecting increased attentional load ($\mu_{\text{multiple}} > \mu_{\text{single}}$ for RT; $\mu_{\text{multiple}} < \mu_{\text{single}}$ for accuracy, one-tailed).

H3 — Modality Effect

H_0: There is no systematic difference in performance between the Lab and Game interfaces ($\mu_{\text{lab}} = \mu_{\text{game}}$).
H_1: Performance differs systematically between the Lab and Game interfaces ($\mu_{\text{lab}} \neq \mu_{\text{game}}$, two-tailed).

H4 — Level Effect

H_0: Performance is stable across game levels, with no linear relationship between level and RT ($r = 0$).
H_1: RT changes systematically across game levels ($r \neq 0$, two-tailed).

H5 — Target Colour Effect

H_0: Target colour has no effect on RT or accuracy ($\mu_{\text{red}} = \mu_{\text{other}}$).
H_1: Red targets result in faster RTs and higher accuracy compared to other colours, consistent with Feature Integration Theory [1] ($\mu_{\text{red}} < \mu_{\text{other}}$ for RT; $\mu_{\text{red}} > \mu_{\text{other}}$ for accuracy, one-tailed).

H6 — Interaction Effect

H_0: The magnitude of the modality penalty does not differ between target load groups.
H_1: The magnitude of the modality penalty differs between the single-target and multiple-target groups (two-tailed).

2 Methods

2.1 Participants and Design

$N = 37$ participants were randomly assigned to one of two Target Load groups. To control for practice and fatigue effects, the order of modalities (Lab first or Game first) was counterbalanced within each group.

Table 1: Participant Group Assignment

Group	**Condition**	**n**	**IDs**
Group A	Single Target	21	1–21
Group B	Multiple Target	16	22–37

2.2 Measures

- **Reaction Time (RT):** Reaction Time (RT) per participant, in seconds. For laboratory trials the raw onset-to-response For game trials, millisecond timestamps were converted to seconds.
- **Accuracy:** Proportion of trials on which the participant correctly identified the target.

2.3 Assumption Checks and Data Transformation

2.3.1 Normality

Shapiro–Wilk tests applied to raw RT means revealed violations of normality in several condition cells. To restore approximate normality a natural-log transformation was applied:

$$\log \mathrm{RT} = \ln(\mathrm{RT}_{\mathrm{mean}})$$

Post-transformation Shapiro–Wilk tests revealed that while normality was restored for the multiple target load ($p = .296$), the single target load condition still departed significantly from normality ($p = .0045$).

2.3.2 Homogeneity of Variance

Levene's test applied to RT across condition cells returned a significant result ($p < .001$), indicating that the assumption of homogeneity of variance was violated.

2.4 Statistical Analyses

1. **Mixed ANOVA (RT):** A 2×2 mixed ANOVA on raw (untransformed) RT tested the main effects of target load and modality, and their interaction. η_p^2 is reported as the effect-size estimate.

2. **Post-hoc paired t-tests:** Following a significant interaction, paired t-tests compared lab and game RT within each target-load group separately, with Cohen's d as the effect size.

3. **Non-parametric corroboration:** Mann–Whitney U tests (between-subjects comparisons) and Wilcoxon signed-rank tests (within-subjects comparisons) were used to corroborate the ANOVA findings.

4. **Accuracy tests:** Mann–Whitney U and Wilcoxon signed-rank tests assessed load and modality effects on accuracy, given that proportions rarely satisfy normality assumptions.

5. **Concurrent Validity (H1):** Pearson correlation between per-participant lab RT and game RT.

6. **Level Effect (H4):** Pearson correlation between game-level number and mean RT across levels.

7. **Colour Effect (H5):** Mann–Whitney U test comparing participant-mean RT for red vs. non-red targets in the lab modality.

All tests were conducted two-tailed at $\alpha = .05$. Bayes Factors (BF_{10}) are reported alongside frequentist p-values where available, providing a continuous measure of evidential strength.

3 Results

3.1 Descriptive Statistics

Table 2: Descriptive statistics for RT and accuracy by condition. Values are means with standard deviations in parentheses.

Target Load	Modality	RT Mean (s)	RT SD (s)	Accuracy (prop.)
Single	Lab	1.495	0.225	1.000
Single	Game	3.104	0.625	0.983
Multiple	Lab	1.505	0.268	0.951
Multiple	Game	1.856	0.507	0.964

Inspection of Table 2 reveals that the game RTs exceed lab RTs in both load conditions.

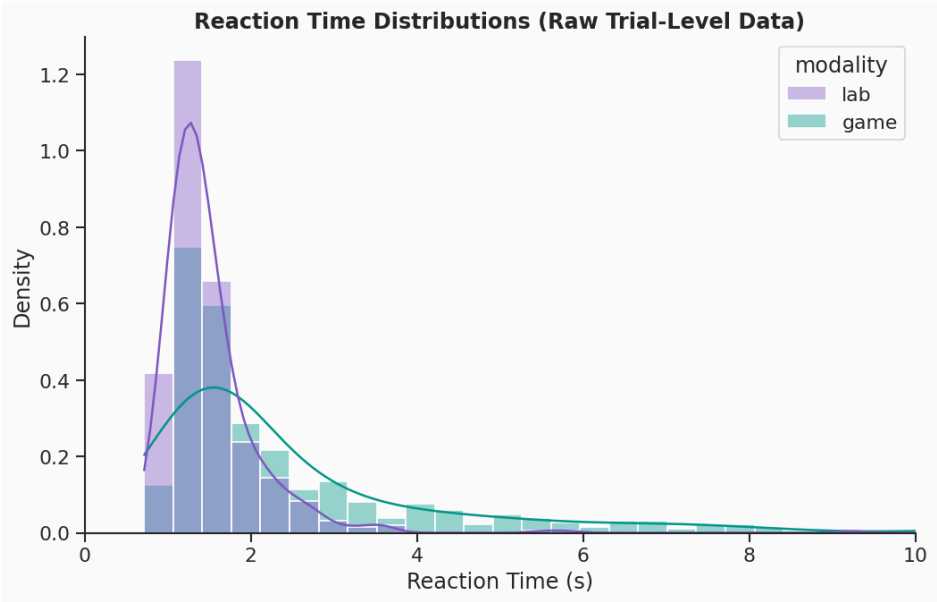

Figure 1: Reaction time distributions (raw trial-level data) for the lab (purple) and game (teal) modalities. The game distribution is substantially right-skewed relative to the lab.

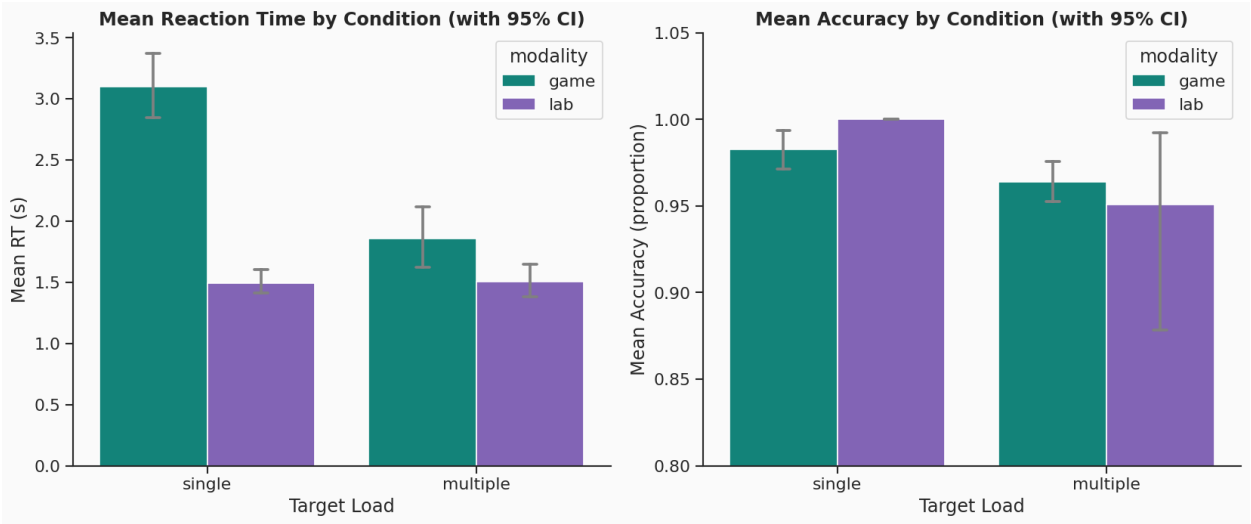

Figure 2: Mean RT (left) and mean accuracy (right) per condition with 95% confidence intervals. Game RTs exceed lab RTs in both target-load conditions; accuracy is lower for multiple targets.

3.2 H2 & H3: Main Effects of Load and Modality

Table 3: 2×2 Mixed ANOVA results for RT. $df_1 = 1$; $df_2 = 35$ for all effects.

Source	F	p	η_p^2	Interpretation
Target Load (H2)	32.23	$< .001$	0.48	Refuted
Modality (H3)	117.97	$< .001$	0.77	Supported
Load $\times$ Modality (H6)	40.34	$< .001$	0.54	Supported

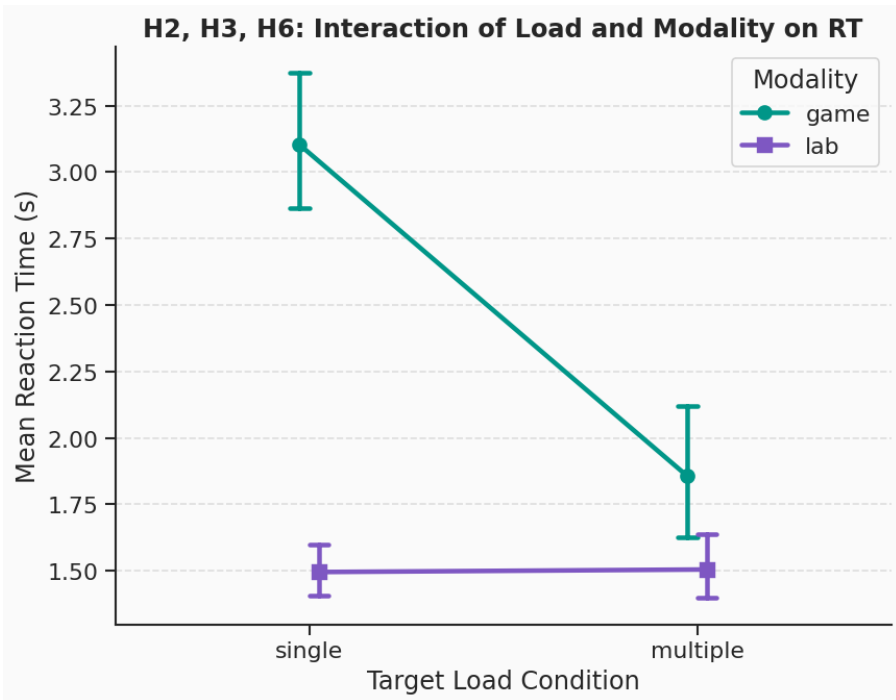

Figure 3: Interaction plot for H2, H3, and H6: mean RT by target load and modality with 95% confidence intervals. Lab RT (purple) remains flat across load conditions, while game RT (teal) drops sharply from single to multiple targets, producing a significant ordinal interaction.

Main effect of target load (H2). $F(1, 35) = 32.23$, $p < .001$. Multiple-target trials were significantly faster than single-target trials, refuting H2.

Main effect of modality (H3). $F(1, 35) = 117.97$, $p < .001$. Game RTs were significantly longer than lab RTs across both load conditions, supporting H3.

3.3 H6: Load $\times$ Modality Interaction

The significant interaction, $F(1, 35) = 40.34$, $p < .001$, indicates that the magnitude of the modality penalty differed between the two load conditions. To unpack this interaction,

separate paired t-tests were conducted comparing lab and game RT within each target-load group.

3.3.1 Post-hoc Paired t-tests

Single-target group. $t(20) = -11.73$, $p < .001$, Cohen's $d = 3.42$, 95% CI $[-1.89, -1.32]$, $\mathrm{BF}_{10} = 4.777 \times 10^7$. The game penalty in the single-target group is extremely large ($d = 3.42$), indicating that participants were dramatically slower on the game platform relative to the lab when tracking only one target.

Multiple-target group. $t(15) = -2.55$, $p = .022$, Cohen's $d = 0.87$, 95% CI $[-0.64, -0.06]$, $\mathrm{BF}_{10} = 2.87$. Although the game penalty remains statistically significant in the multiple-target group, its magnitude is substantially smaller ($d = 0.87$) than in the single-target group.

These results reveal a counterintuitive pattern: the game interface imposes a *proportionally greater penalty* when the task is cognitively simpler (single target). One plausible explanation is that the single-target search is rapid and automatic in the lab (feature search under Feature Integration Theory), whereas the game's unfamiliar interface disrupts this automaticity, disproportionately slowing single-target responses. Under multiple-target load, both lab and game conditions require slower, conjunction-based search, attenuating the relative game penalty.

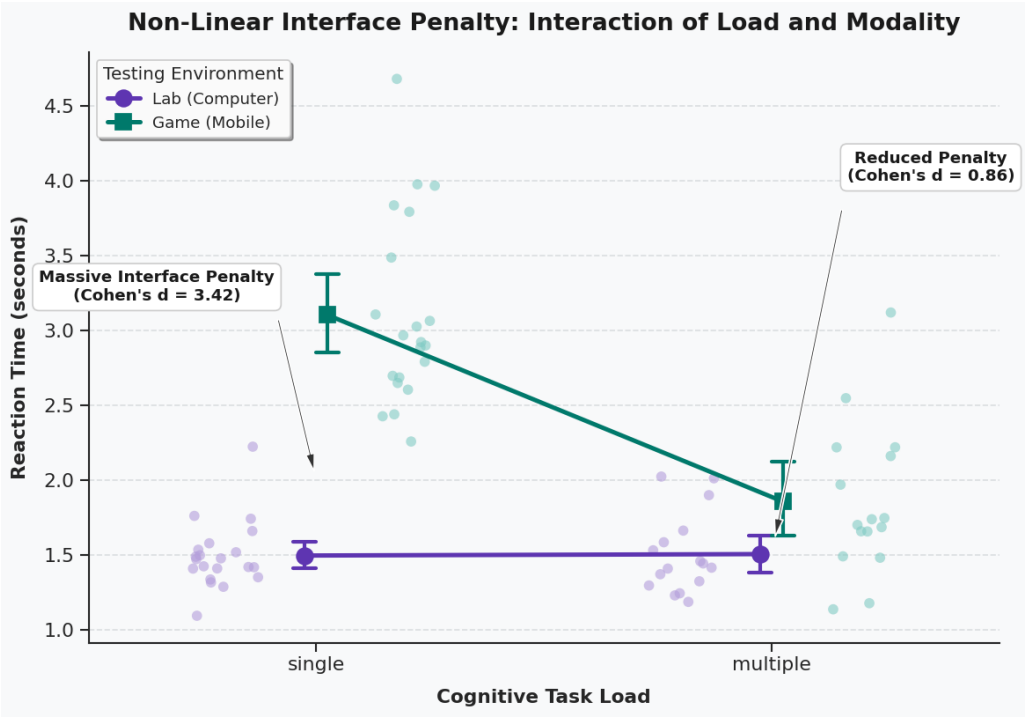

Figure 4: H6: Non-linear interface penalty. Individual participant data points (jittered) with group means and 95% CI. The game penalty is massive in the single-target condition (Cohen's $d = 3.42$) but substantially reduced for multiple targets ($d = 0.87$).

3.4 Non-Parametric Corroboration

To confirm that the ANOVA findings on RTs generalise to raw (untransformed) RT, non-parametric tests were applied to the original data.

3.4.1 Reaction Time

- **Load effect (Mann–Whitney U):** $U = 914.0$, $p = .008$, RBC = 0.36, CLES = 0.68. Corroborates the significant load effect detected by the ANOVA.
- **Modality effect (Wilcoxon signed-rank):** $W = 24.0$, $p < .001$, RBC = -0.93, CLES = 0.11. A large rank-biserial correlation ($|\text{RBC}| = 0.93$) confirms a strong, systematic modality effect consistent with the ANOVA.

3.4.2 Accuracy

- **Load effect on accuracy (Mann–Whitney U):** $U = 1009.0$, $p < .001$, RBC = 0.50, CLES = 0.75. Multiple-target conditions produced significantly lower accuracy.
- **Modality effect on accuracy (Wilcoxon signed-rank):** $W = 50.0$, $p = .014$, RBC = 0.60, CLES = 0.68. Game accuracy was significantly lower than lab accuracy.

3.5 H1: Concurrent Validity

A Pearson correlation between per-participant lab RT and game RT (across all conditions) yielded $r = 0.076$, $p = .656$. This result does *not* support H1. The near-zero correlation indicates that individual differences in lab RT do not predict individual differences in game RT. A possible explanation is that the game interface introduces large, participant-specific learning or adaptation effects that overwhelm any underlying attentional trait shared across platforms.

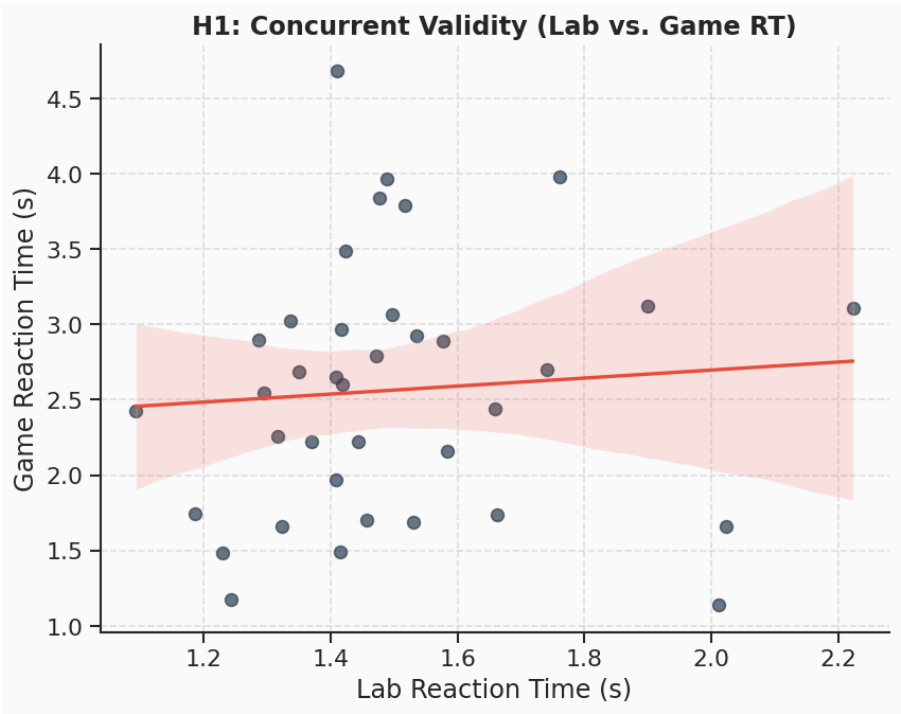

Figure 5: H1: Concurrent validity scatter plot. Each point is one participant's mean lab RT vs. mean game RT. The flat regression line ($r = 0.076$, $p = .656$) confirms no significant concurrent validity relationship between the two platforms.

3.6 H4: Level Effect

Within the game modality, the Pearson correlation between level number and mean RT across levels was $r = 0.802$, $p < .001$. This strong positive correlation supports H4: reaction time increased steadily across sequential game levels, consistent with fatigue accumulation or rising task difficulty as levels progressed.

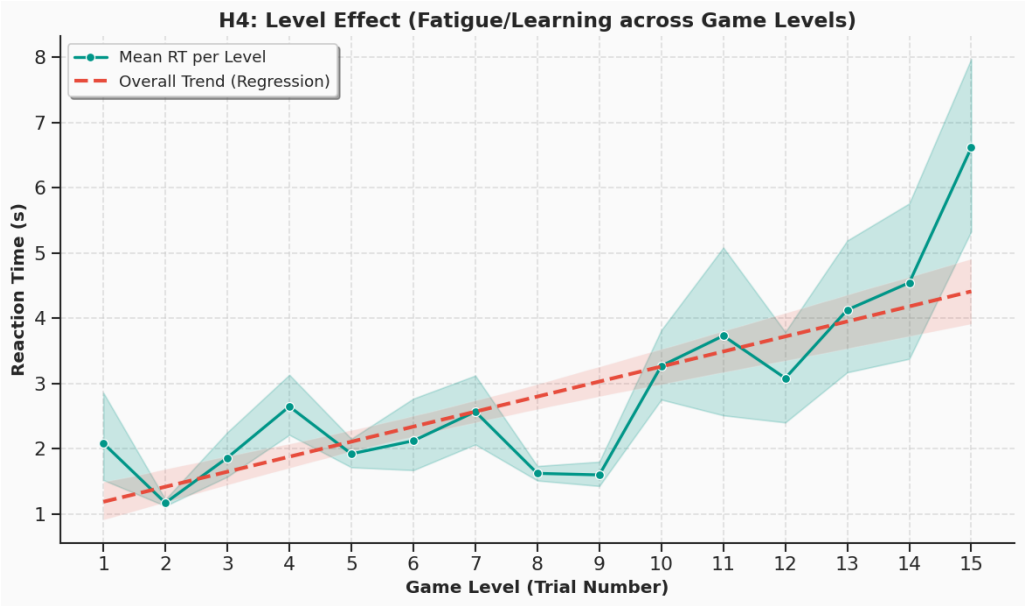

Figure 6: H4: Level effect in the game modality. Mean RT per level (teal line with 95% CI shading) rises across 15 game levels. The dashed red regression line confirms a strong positive trend ($r = 0.802$, $p < .001$).

3.7 H5: Target Colour Effect

A Mann–Whitney U test comparing participant-mean RT for red versus non-red targets in the lab modality returned $U = 395.0$, $p = .002$, RBC $= -0.42$. Red targets were associated with faster responses than non-red targets, supporting H5 and consistent with the colour-salience prediction from Feature Integration Theory: a uniquely coloured target "pops out" from a display of differently coloured distractors, enabling faster feature search.

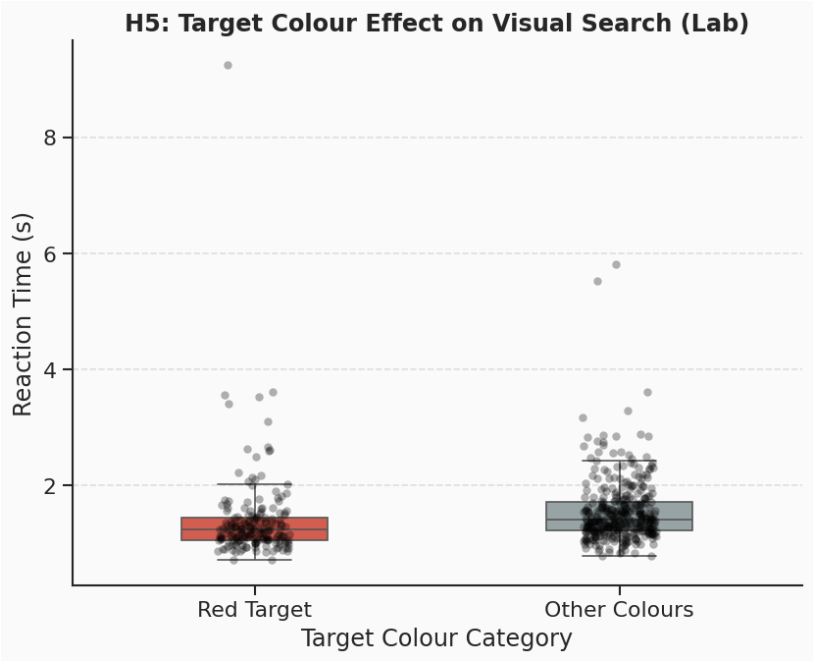

Figure 7: H5: Target colour effect in the lab modality. Red targets (left) produced faster RTs than other-coloured targets (right), consistent with a colour pop-out (feature search) advantage ($U = 395.0$, $p = .002$).

4 Conclusion

Table 4 provides a consolidated overview of all hypothesis tests and their outcomes.

Table 4: Summary of hypothesis tests and conclusions.

Hypothesis	**Test**	**Result**	**Conclusion**
H1: Concurrent Validity	Pearson r	$r = 0.076$, $p = .656$	**Not supported.** No significant correlation.
H2: Load Effect	2 × 2 Mixed ANOVA	$F(1, 35) = 32.23$, $p < .001$	**Refuted.** Multiple targets resulted in faster RTs.
H3: Modality Effect	2 × 2 Mixed ANOVA	$F(1, 35) = 117.97$, $p < .001$	**Supported.** Game interface slows RT.
H4: Level Effect	Pearson r	$r = 0.802$, $p < .001$	**Supported.** RT rises across game levels.
H5: Target Colour	Mann–Whitney U	$U = 395.0$, $p = .002$	**Supported.** Red targets detected faster.
H6: Interaction Effect	2 × 2 Mixed ANOVA	$F(1, 35) = 40.34$, $p < .001$	**Supported.** Game penalty is disproportionately larger for single-target ($d = 3.42$) than for multiple-target ($d = 0.87$) conditions.

This Phase 2 analysis confirms significant modality (H3) and interaction (H6) effects: the game interface systematically slows reaction times, particularly during single-target feature search. However, the load effect hypothesis (H2) was refuted, as multiple-target trials unexpectedly yielded faster responses. Non-parametric tests corroborated these findings. Furthermore, target colour salience (H5) and game-level fatigue (H4) significantly impacted performance, serving as important methodological cautions.

Critically, lab and game performance were uncorrelated, failing to demonstrate concurrent validity (H1). This challenges the game's viability as a direct drop-in replacement for laboratory measures. Future research should explore whether familiarisation training, adaptive levelling, or interface redesigns can bridge this validity gap.

References

[1] Anne M. Treisman, Garry Gelade (1980). A feature-integration theory of attention. *Cognitive Psychology*, Volume 12, Issue 1.

[2] Lumsden et al. (2016). Gamification of cognitive assessment and training: A systematic review of applications and efficacy. *JMIR Serious Games*.